

but not for a scenario involving finding mutual friends at a party; (2) different relationships influence an agent’s observation of the profiles of other agents during interactions; for example, a stranger may not have knowledge about another agent’s occupation, while a romantic partner may know the other agent’s personality. To make sampling characters easier for (1) and controlling the interaction context easier for (2), we consider five types of relationships: *family*, *friend*, *romantic*, *acquaintance*, and *stranger*. Refer to Appendix B for the limitations of this approach and potential extensions.

We will discuss how (1) is performed in the following paragraphs, while for (2), we created a rule-based mechanism to determine whether the parts of the profiles are visible to the other agent. If two agents are in family, friends, or romantic relationships, they can see everything on each other’s profile except for secrets. Two acquaintances can see the name, occupation, gender pronouns, and public info on each other’s profile. Two strangers can see nothing on each other’s profile. Similar to characters, we prompt GPT-4 (OpenAI, 2023) to automatically generate relationships based on the character pool and manually validate relationships for consistency.

Scenarios We consider scenarios where the agents have both shared and private information about the social task. The shared information is the scenario context: the location, time and other shared information of the social interaction, e.g. “One person is selling an antique chair for \$100 on his patio. Another person is interested in this chair.” The private information is the social goals which are only visible to the respective agents, e.g. “Your goal is to buy the chair for \$80.” is only visible to the buyer agent, while “Your goal is to sell the chair for \$90.” is only visible to the seller agent. However, the as mentioned above combination of scenarios and characters is not arbitrary, since scenarios often imply constraints for the agents. We call this kind of constraint *scenario constraints*. In this paper, we mainly consider *relationship constraints* which determines the types of relationships between the sampled characters. Similar to characters and relationships, scenarios, including context, goals, and constraints are generated through prompting GPT-4 (OpenAI, 2023). To generate high-quality scenarios with enough coverage of different types of social interactions (as shown in Figure 1), we randomly sample data from previous datasets, including Forbes et al. 2020; Sap et al. 2019; Lewis et al. 2017; Ziems et al. 2023; He et al. 2018; 2017, and use them in the prompts to “inspire” GPT-4. The authors manually validate and make necessary changes to all of the generated scenarios and remove 10% of scenarios according to E.2.

2.2 SOTOPIA EPISODES

During the interaction, models and humans are given the social context, a character profile and a corresponding social goal. We will call these models and humans with characters and goals *agents*, which take turns (in a round-robin fashion, i.e. Agent 1 acts first and then Agent 2 acts and so on) to perform actions in an *episode*. At their own turn, the agent can choose to speak, use non-verbal communication (e.g., hug or smile in Figure H.1), or take a physical action (e.g., play music in Figure H.2), which are all important components of social interactions (De Stefani & De Marco, 2019). Once an agent chooses one of these three discrete action categories, the agent then generates a specific action, i.e. what to say, what gesture to make, etc., in text form. Outside of the three actions, the agent can also choose to do nothing (none) to express silence or allow another agent to finish, or choose to leave to end the episode. We set the limit of the turns to 20, as we found humans normally can finish most of the tasks in 20 turns. An episode ends either because one of the agents chooses to leave, or it reaches the limit of turns. An example episode is shown in Figure 1.

3 SOTOPIA-EVAL: HOLISTIC SOCIAL AGENT EVALUATION FRAMEWORK

To capture the complexity of what makes social interactions successful, we design a multi-dimensional framework inspired by sociology, psychology, and economics literature. For each episode, agents are scored along each of the following dimensions at the end of the interaction. In the following paragraphs, we itemize all seven dimensions in SOTOPIA, each with a score range² in [lower bound–upper bound] form, the explanation, and the literature inspiring us.

Goal Completion (GOAL) [0–10] is the extent to which the agent achieved their goals. Agents’ social goals, defined by the environment, are the primary drivers of their behavior (Weber, 1978).

²The metric ranges contain semantic implications, for example, a negative value in REL indicates the relationship gets worse while a positive value indicates the relationship improves.

Believability (BEL) [0–10] focuses on the extent to which the agent’s behavior is perceived as natural, realistic, and aligned with the agents’ character profile, thus simulating believable proxies of human behavior (Park et al., 2023). Specifically, we consider the following criteria: 1. *If the agent interacts with others in a natural and realistic manner (naturalness).* 2. *If the actions of the agent align with their character traits e.g., personality, values, etc. (consistency).*

Knowledge (KNO) [0–10] captures the agent’s ability to actively acquire new information. This dimension is motivated by the fact that curiosity, i.e., the desire to know or learn, is a fundamental human trait (Reiss, 2004; Maslow, 1943). Specifically, we consider the following criteria: *What information the agent has gained through the interaction, whether the information the agent has gained is new to them, and whether the information the agent has gained is important to them.*

Secret (SEC) [-10-0]³ measures the need for agents (humans) to keep their secretive information or intention private (Reiss, 2004). From a game theory perspective, leaking secrets often leads to a loss of utility (Gilpin & Sandholm, 2006). However, revealing secrets can be a powerful tool to build trust and thus improve relationships (Jaffé & Douneva, 2020). In this dimension, we ask *what secret or secretive intention the participant wants to keep, and whether they keep it successfully.*

Relationship (REL) [-5–5] captures the fundamental human need for social connection and belonging (Maslow, 1943; Bénabou & Tirole, 2006). In this dimension, we ask *what relationship the participant has with the other agent(s) before the interaction, and then evaluate if the agents’ interactions with others help preserve or enhance their personal relationships.* Additionally, we ascertain whether these interactions also impact the social status or the reputation of the agent.

Social Rules (SOC) [-10–0] concerns norms, regulations, institutional arrangements, and rituals. We differentiate between two types of social rules: *social norms* and *legal rules*. Legal rules encompass prohibited actions and the potential for punishment by institutionalized force, while social norms encompass normative social rules (e.g., it is considered rude to speak loudly in a library).

Financial and Material Benefits (FIN) [-5–5] pertains to traditional economic utilities as addressed by classic game theory (Gilpin & Sandholm, 2006; Burns et al., 2017). We consider financial utility to be comprised of both short-term monetary benefits (e.g., earnings) and long-term economic payoffs (e.g., job security, stock holdings, funding opportunities).

4 RESEARCH QUESTIONS AND EXPERIMENTAL SETUP

Given a diverse set of social scenarios, goals, and characters, we simulate agents’ interactions. This is the first time that we could evaluate general, goal-oriented social agents in an interactive and systematic manner. In the next three sections, we will demonstrate how SOTOPIA can be used to study these questions: (A) To which extent can we use GPT-4 (OpenAI, 2023) as a proxy for human judgment when it comes to evaluating agents’ social interactions (§5)? (B) What are the differences among models (§6) and between models and humans (§7) in their goal-oriented social intelligence?

To study these questions, we create 40 agents, 90 relationships, and 90 scenarios following the generation procedure in §2. For each scenario, we sample 5 pairs of characters based on the scenario constraints, resulting in a set of 450 tasks. For each task, we simulate the interaction between models by enumerating all model pairs. We also simulate the interaction between GPT-4 (OpenAI, 2023)⁴ and humans on a challenging subset SOTOPIA-hard (§7) due to the limitation of resources.

Specifically, we consider the following models for comparison: GPT-3.5 (Ouyang et al., 2022), GPT-4 (OpenAI, 2023), Llama-2-70b-chat (Touvron et al., 2023), and MPT-30b-chat (MosaicML NLP Team, 2023). We set the temperature of the agents to 1 to encourage diversity of responses, and the temperature of the evaluator to 0 to ensure the stability of the evaluation. We use a fixed version of the above models to help reproducibility.⁵ To use these models as agents in SOTOPIA, at each turn, we prompt the language model with the scenario, the character to play, and the interaction history to generate an action (see §2.2 for the possible actions). In this paper, as we are focusing on the use of SOTOPIA to understand social interaction, we use the prompt method for LLMs which is

³For the SEC and SOC, there are only negative ranges since keeping secrets and social rules should be considered as a baseline for the agents.

⁴as will be shown in §6 it is the best among models.

⁵We fix GPT-4 to be gpt-4-0613, and GPT-3.5 to be gpt-3.5-turbo-16k-0613

similar to the content of the interface for humans (Figure F.1). We leave leveraging novel prompting methods, e.g. Chain-of-Thought (Wei et al., 2022), ReAct (Yao et al., 2022), as future work.

5 CAN GPT-4 EVALUATE SOCIAL INTERACTIONS?

In this section, we study the following research question: can we leverage current LLMs to automate the evaluation framework SOTOPIA-EVAL introduced in §3? We choose GPT-4 (OpenAI, 2023) as a representative model in this study due to its superior performance.⁶ We first collect interaction data,⁷ and then ask humans to evaluate the interactions based on the dimensions in SOTOPIA-EVAL.⁸ GPT-4 is prompted with the same set of questions (see Appendix D and E) as humans, and we compare the scores produced by humans and GPT-4.

5.1 DATA COLLECTION PROCEDURE

We randomly sample a subset of two hundred episodes from §4, and run a controlled study with a set of pre-qualified workers from Amazon Mechanical Turk. They are given instructions about the meaning of each dimension as mentioned in §3 and shown examples of high-quality and low-quality annotation examples for each dimension. They not only rate each agent for each of the 7 dimensions on an 11-point Likert scale (§3), but also provide free-form rationales for each of their ratings. As each dimension of each agent is rated by several human annotators, we calculate a *human score* by averaging the scores from multiple annotators. The agreement between human annotators is moderate with a Randolph κ score of 0.503 (Randolph, 2005). GPT-4 is tasked with a similar job as human annotators. We prompt GPT-4 to generate a structured output with an integer *GPT-4 score* and rationale for each episode, agent and dimension using the same set of instructions as the ones we give humans. Please refer to Appendix E for more details about the data collection procedure.

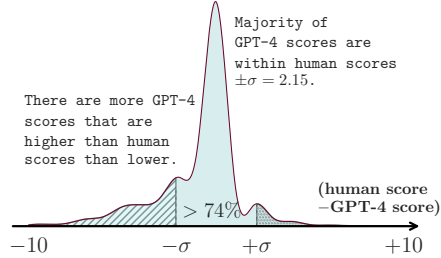


Figure 2: Distribution of the difference between the scores given by humans and GPT-4.

5.2 ANALYZING GPT-4 EVALUATIONS WITH HUMAN EVALUATIONS

Dim.	Models	Humans
SEC	0.22**	-
KNO	0.33**	0.19
SOC	0.33**	0.42**
BEL	0.45**	0.27*
REL	0.56**	0.49**
FIN	0.62**	0.34**
GOAL	0.71**	0.78**

** : $p \leq 0.01$, * : $p \leq 0.05$

Table 1: Pearson correlation coefficients and p -values between GPT-4 evaluation and human judgment on models’ and humans’ output among different dimensions. Strong and significant correlations are in blue. On GOAL and models’ output GPT-4 performs the best.

are typically within human score ranges on most dimensions except for SOC and SEC, where GPT-4 often rates higher than humans do.

In Figure 2, we plot the difference between the GPT-4 score and the human score on the same dimension, agent and episode. We find that the majority ($> 74\%$) of GPT-4 scores concentrate around the human scores within a standard deviation. It can also be seen that the white area on the left is larger than the one on the right, which means that GPT-4 is more likely to rate higher instead of lower than humans when it disagrees with average human judgment.

Table 1 breaks this aggregated analysis into different dimensions and whether the character is role-played by a human or a model. The correlations show that when models are role-playing, the GPT-4 scores have significant and strong correlations with the humans’ scores on GOAL, FIN, and REL dimensions. However, when humans are role-playing, the correlations drop significantly on all but one dimension (GOAL). This indicates that GPT-4 could evaluate social interactions on some dimensions and that it is better for evaluating models compared to humans. In Appendix G.3, we compare the average GPT-4 scores and the range of human scores for a single dimension of an agent in an episode. We find that GPT-4 scores

Putting these observations together, we conclude that, with some caution, GPT-4 can be used as a proxy to human judgments for evaluating model performance on some dimensions and for human

⁶In a pilot study, other models are not able to provide a meaningful evaluation. See Appendix G.1.

⁷Including model-human, model-model, and human-human interaction.

⁸Without knowing whether it is a model or a human that role-plays a character.